

MORELIA INTL, Mexico

WMO: 766655

Lat: 19.850N

Lon: 101.026W

Elev: 1839

StdP: 81.10

Time Zone: -6.00 (MEX)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	2.8	4.0	-5.1	3.1	17.1	-3.1	3.7	17.4	9.8	19.4	8.5	19.8	0.0	180	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	16.7	31.9	15.7	30.4	15.2	29.2	15.0	18.7	25.2	18.3	24.7	18.0	24.5	3.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	15.3	19.7	16.8	15.0	19.5	16.1	14.4	19.1	61.7	25.7	59.8	24.2	58.8	24.6	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.6	6.1	-0.5	34.1	1.9	1.3	-1.9	35.0	-3.0	35.8	-4.1	36.5	-5.5	37.5		
			-1.8	20.1	1.9	1.3	-3.2	21.0	-4.3	21.8	-5.4	22.5	-6.8	23.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	14.3	16.1	17.7	19.9	21.8	21.4	19.9	19.7	19.4	18.2	15.7	14.4
	DBStd	2.94	1.58	1.65	1.87	1.60	1.66	1.69	1.14	1.07	1.10	1.73	1.87	1.59
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	461	126	65	32	4	1	1	2	2	4	22	81	123
	CDD10.0	2998	133	170	238	298	366	342	307	300	283	256	171	136
	CDD18.3	416	0	2	11	52	108	93	50	43	36	19	1	0
	CDH23.3	5102	74	261	558	1012	1242	746	278	256	218	243	139	74
	CDH26.7	1236	1	27	113	322	472	236	22	16	8	15	5	1
Wind	WSAvg	1.7	1.6	1.8	2.1	2.1	2.0	1.9	1.4	1.4	1.4	1.6	1.4	1.5
Precipitation	PrecAvg	772	17	7	7	14	42	153	164	156	138	55	13	5
	PrecMax	969	112	30	41	41	75	315	242	215	238	92	39	26
	PrecMin	476	0	0	1	0	0	49	89	66	52	14	0	0
	PrecStd	133	26	8	10	12	24	56	43	43	52	27	11	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.2	29.0	30.2	32.1	33.8	33.2	28.2	28.1	27.8	28.1	27.2	26.2
		MCWB	13.1	13.7	14.1	15.1	16.5	16.6	17.4	17.7	17.2	16.6	15.1	14.0
	2%	DB	25.1	27.2	29.1	31.0	32.0	31.1	27.1	27.0	26.8	27.0	26.0	25.1
		MCWB	13.0	13.4	13.9	14.8	15.9	16.1	17.6	17.6	17.3	16.5	14.7	13.7
	5%	DB	24.1	26.2	28.0	30.0	30.9	29.8	26.1	25.9	25.8	25.9	25.0	24.1
		MCWB	12.7	13.3	13.7	14.4	15.6	16.3	17.5	17.4	17.2	16.2	14.5	13.5
	10%	DB	23.0	25.0	26.8	28.9	29.8	27.9	25.0	24.9	24.8	24.9	23.9	23.0
		MCWB	12.5	13.1	13.5	14.1	15.4	16.5	17.4	17.6	17.4	16.2	14.5	13.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.1	15.7	16.2	17.1	18.4	19.1	19.2	19.2	19.2	18.7	17.5	16.4
		MCDB	23.9	26.0	26.6	28.4	27.9	26.7	25.5	25.5	25.0	25.0	24.5	24.1
	2%	WB	14.2	14.9	15.4	16.1	17.7	18.3	18.4	18.5	18.4	18.1	16.5	15.1
		MCDB	22.9	25.0	25.9	27.8	27.0	25.6	24.6	24.6	24.2	24.2	23.3	22.6
	5%	WB	13.5	14.1	14.8	15.4	17.1	18.0	18.1	18.1	18.1	17.5	15.7	14.2
		MCDB	22.2	24.0	25.5	27.2	26.6	25.1	24.3	24.1	23.9	23.4	22.5	21.8
	10%	WB	12.9	13.5	14.1	14.8	16.7	17.5	17.6	17.7	17.6	17.0	15.1	13.6
		MCDB	21.5	23.3	24.7	26.7	26.1	24.1	23.6	23.4	23.0	22.6	21.8	21.2
Mean Daily Temperature Range	5% DB	MDBR	17.8	18.1	18.8	18.8	16.7	13.3	11.8	11.8	12.0	14.4	17.5	18.2
		MCDBR	19.8	20.5	20.7	19.7	18.2	16.3	13.2	13.3	13.6	16.0	18.7	19.8
		MCWBR	9.6	9.1	8.8	7.6	6.4	5.4	5.2	5.5	5.7	7.0	8.9	9.8
	5% WB	MCDBR	17.6	18.0	18.3	17.8	15.8	13.1	11.9	11.9	11.7	13.4	16.4	17.9
		MCWBR	8.9	8.4	8.1	7.3	5.6	4.7	5.1	5.2	5.2	6.1	8.3	9.5
Clear-Sky Solar Irradiance	taub	0.329	0.339	0.336	0.375	0.452	0.413	0.400	0.394	0.397	0.372	0.331	0.326	
	taud	2.434	2.400	2.406	2.312	2.123	2.270	2.317	2.331	2.328	2.395	2.464	2.468	
	Ebn at Noon	950	958	971	928	849	875	887	898	897	911	941	942	
	Edh at Noon	106	116	120	133	160	137	131	129	128	115	102	100	
All-Sky Solar Radiation	RadAvg	4.86	5.76	6.68	7.15	6.64	5.85	5.71	5.71	5.16	5.21	5.02	4.74	
	RadStd	0.35	0.52	0.37	0.26	0.39	0.38	0.36	0.30	0.33	0.34	0.37	0.26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Regional (5 neighbors)	N/A	+0.80	-1.73	N/A	-0.74	-0.94	-30	N/A	N/A		

Nomenclature: See separate page